

PROJECT DELIVERY DASHBOARD

Software Requirements Specification (SRS) & System Architecture

Project Name:	Project Delivery Dashboard	Version:	1.0.0 (Production Candidate)
AI Engine:	Groq API (Llama 3.3 70B)	Date:	August 2026
Frontend:	React + TypeScript + Vite	Backend:	Python FastAPI + SQLite

1. Executive Summary & Problem Statement

The Challenge: Customer delivery operations across onboarding, hardware integrations, and milestone deliverables are frequently tracked internally across siloed channels (emails, Slack threads, client calls). Status updates become scattered, leading to visibility bottlenecks, human overhead, and misaligned client expectations.

The Solution: A unified Project Delivery Dashboard powered by the ultra-low latency **Groq API (Llama 3.3 70B)** that automatically synthesizes unstructured communications into structured milestone statuses while strictly isolating internal technical blockers from customer-safe client portals.

2. Scope & Dual-Persona Architecture

The system enforces strict view-layer segregation across two core user roles:

- **Internal Delivery Team View:** Full operational control. Displays raw technical blocker causes, root-cause analyses, developer assignment matrices, activity feeds, and real-time Linear issue webhook synchronizations.
- **Customer Client View:** Sanitized, executive-level delivery portal. Obfuscates internal friction and credentials issues, displaying verified progress percentages, high-level milestone timelines, and professional status updates.

3. Functional Requirements (FR)

Req ID	Module	Requirement Description	Priority
FR-01	Project Portfolio	Display multi-project portfolio with health status (ON_TRACK, AT_RISK, BLOCKED, COMPLETED), multi-owner avatars, progress rings, and target delivery dates.	Critical
FR-02	Milestones & Tasks	Interactive milestone progression pipeline supporting Open, In Progress, Blocked, and Done states with automatic completion recalculations.	Critical
FR-03	Groq AI Ingestion	Process unstructured emails, Slack messages, and call transcripts via Groq LLM API. Extract structured JSON updates, task statuses, blockers, and customer summaries.	Critical
FR-04	Dual-View Isolation	Toggle between Internal View and Customer View with client tenant isolation ensuring no confidential internal notes leak to customers.	Critical
FR-05	NL Query Assistant	Conversational portfolio query assistant powered by Groq LLM to answer delivery status questions and highlight relevant projects.	High
FR-06	Issues Management	Track project issues with categories (Bug, Feature Request, Question, Support, Implementation) and internal-only access tags.	High
FR-07	Linear Integration	Simulate and receive Linear webhook payloads to automatically update project tasks and recalculate overall progress.	Medium
FR-08	Stale Detection	Automatically detect and trigger banner alerts for stale projects with no movement for > 7 days.	Medium

4. AI Engine & Groq LLM Integration

Primary LLM Engine: Official Groq Python SDK (groq>=0.9.0) with model llama-3.3-70b-versatile utilizing JSON mode (response_format={'type': 'json_object'}) for sub-second structured entity extraction.

Multi-Tier Fallback Cascade:

- Tier 1 (Groq API):** Lightning-fast cloud inference with Llama 3.3 70B (Primary).
- Tier 2 (Google Gemini API):** Secondary cloud LLM fallback.
- Tier 3 (Local NLP Engine):** Zero-dependency offline fallback using Scikit-Learn TF-IDF semantic task matching and sentiment regex classification.

5. Non-Functional Requirements (NFR)

- **Performance:** AI entity extraction response time < 1.5 seconds via Groq hardware acceleration. Frontend bundle load time < 300ms.
- **Security & Privacy:** Zero credentials leakage. API keys stored strictly in backend/.env and excluded from source control via .gitignore.
- **Reliability:** Offline resilience ensuring the application operates seamlessly even without active external API access through local NLP.
- **Compatibility:** Fully responsive modern glassmorphic interface supporting dark and light themes across modern desktop browsers.

6. Verification & Quality Assurance

Test Suite	Target Component	Result
FastAPI Endpoints Suite	All 9 REST endpoints (/api/health, /api/projects, /api/nl-query, etc.)	100% PASS (9/9)
Groq Live Extraction Test	Multi-channel unstructured text parsing & JSON formatting	100% PASS
NL Query Assistant Test	Portfolio-wide question answering and project ID filtering	100% PASS
TypeScript & Vite Build	Frontend compilation and type verification	0 Errors (Clean)